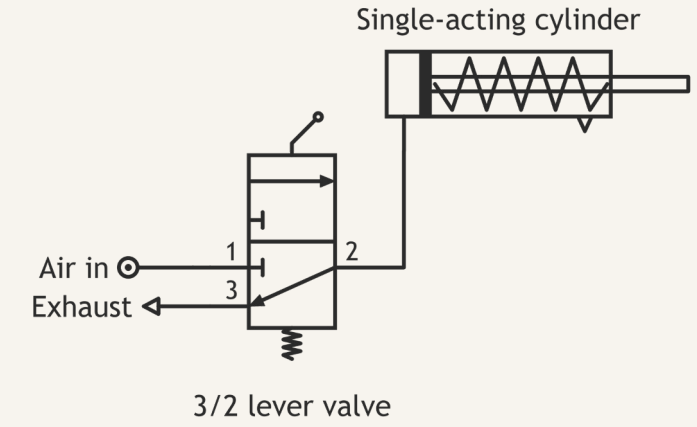
Daily Review

1	**State** the **regulator** setting to **change** a tube. (**value and units**)	0 bar
2	**State** the **regulator** setting for **testing**. (**values and units**)	1 to 2 bar
3	**State** where a circuit is **tested first**.	In the simulation
4	**State** the **three parts** of a system.	Input, process, output
5	**Name** the energy of **movement**.	Kinetic energy

Reading the Diagram

Build the lever gate. Then draw it.



S3 Engineering Science

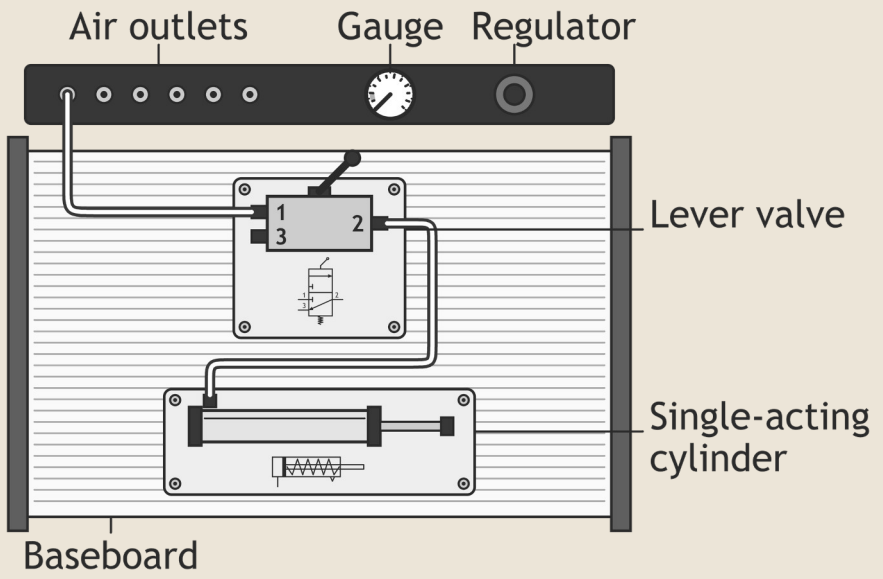
The Job: The Lever Gate

i When the lever is pushed, the rod moves out.

ii When the lever is let go, the rod moves back in.

iii The drawing shows every symbol and port.

Predict where the **air** goes when you **let go** of the lever.



By the end of this lesson…



1. I can say what a **3/2 valve** does when it is **pressed** and **let go**
2. I can **draw** a circuit with **symbols**
3. I can **build** a circuit from a **bench card**

THE BUILD

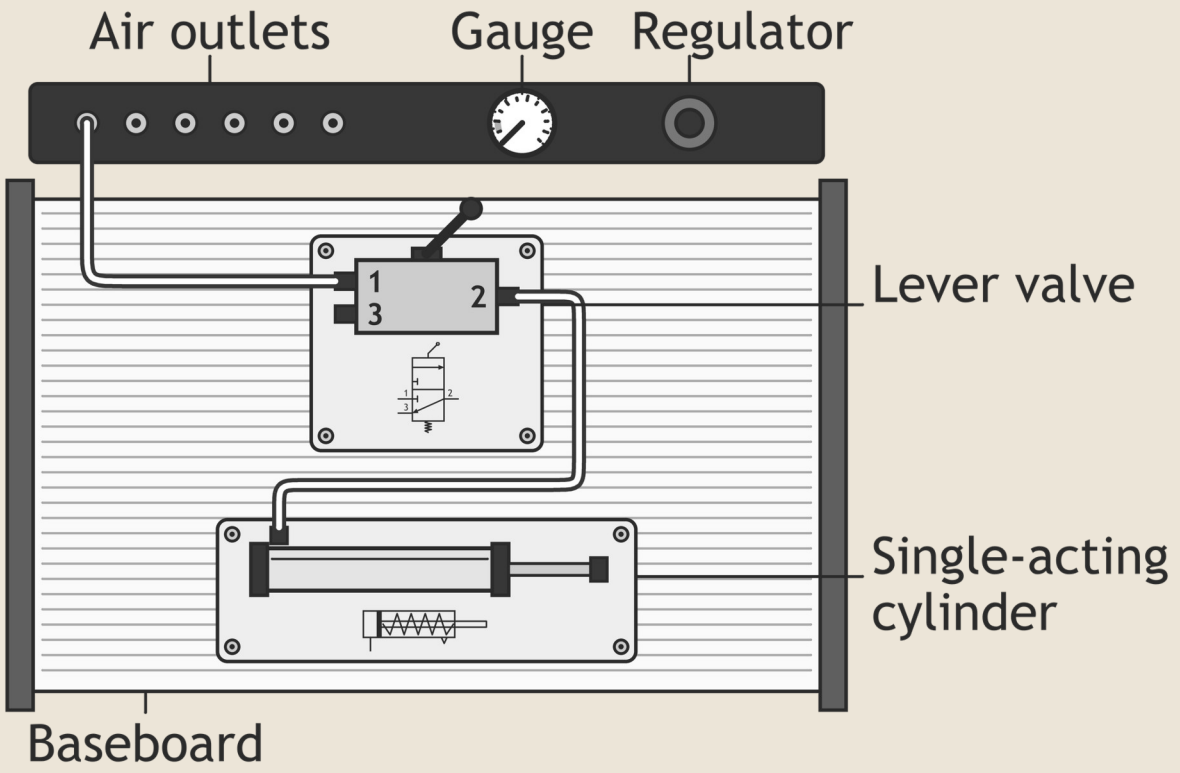
The Lever Gate

Simulate it. Air Permit. Then the kit.

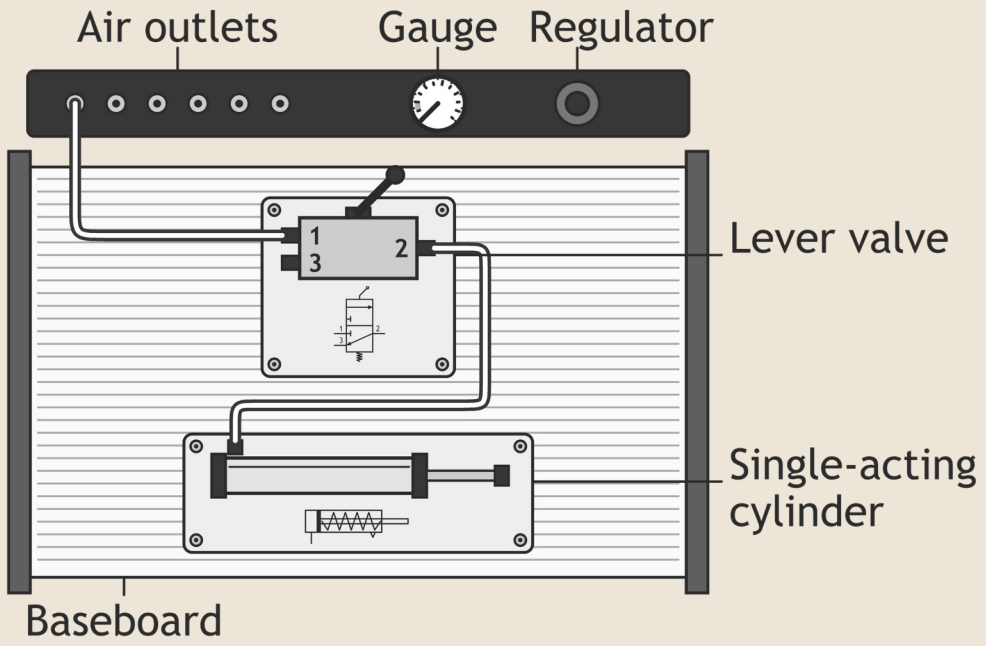
What to Collect

Builder and Safety: one set per team.

1 Bench card 2
2 Manifold (air supply)
3 3/2 lever valve
4 Single-acting cylinder
5 Tubing
6 Goggles, one each



Simulate and Build

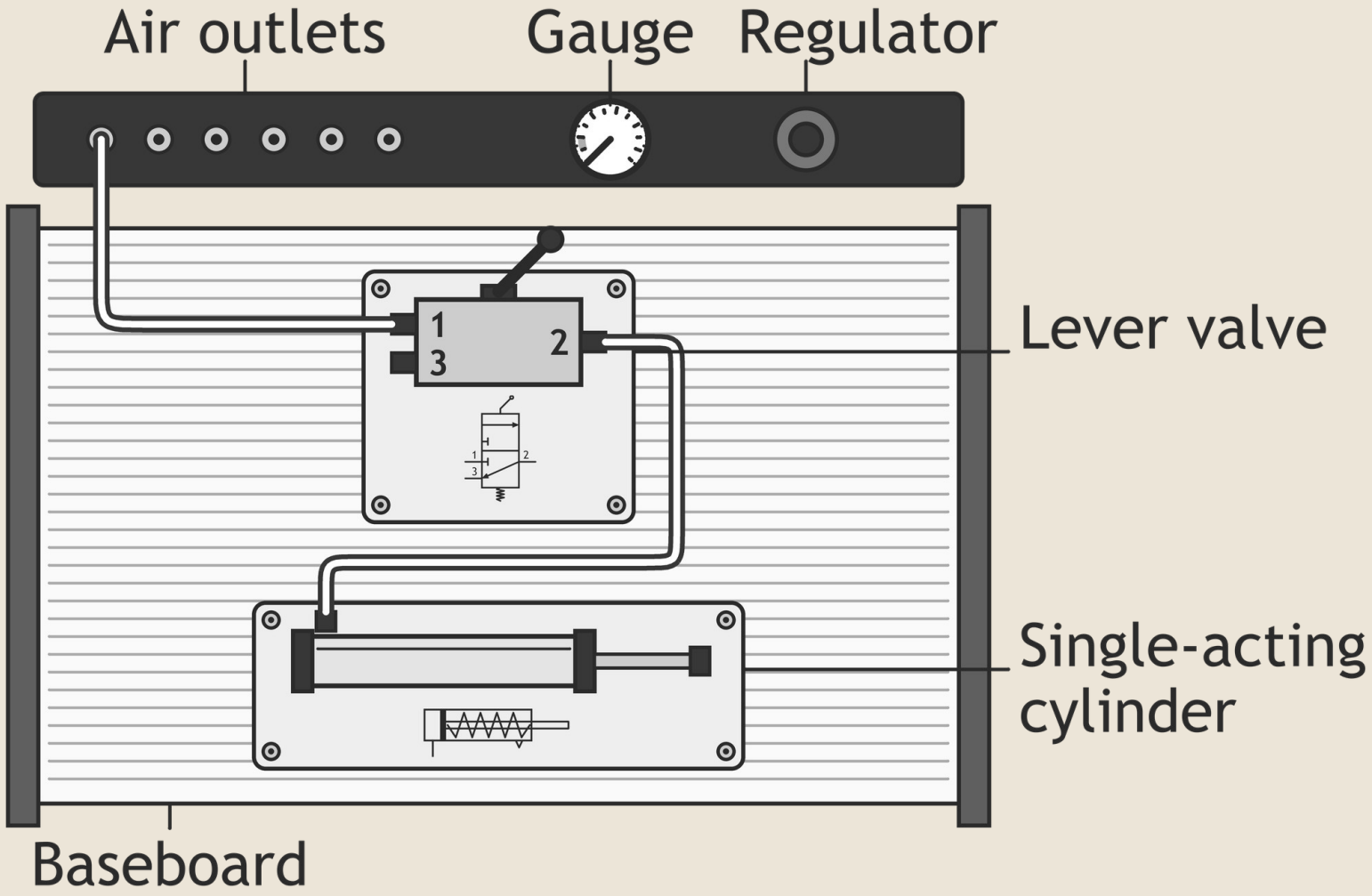


SAFETY Regulator at 0 to build. 1 to 2 bar to test.

Tasks 3-4 (B6A-L2-p6): **Simulate**, then **build** it.
Check the Air Permit. Push, let go, **listen**.

10 min

Build It: Bench Card 2



Test It: What Did You Find?

1. **State** whether **point i** was met, and what **pushed** the rod **out**.

   Yes: air from the valve

2. **State** whether **point ii** was met, and what **pushed** the rod **back in**.

   Yes: the cylinder's spring

3. **State** where the **air** went when you **let go** of the lever.

   Out of the valve's exhaust, port 3

Parts of the Symbol (1 of 2): Boxes and Ports

Box

one state of the valve

2 boxes

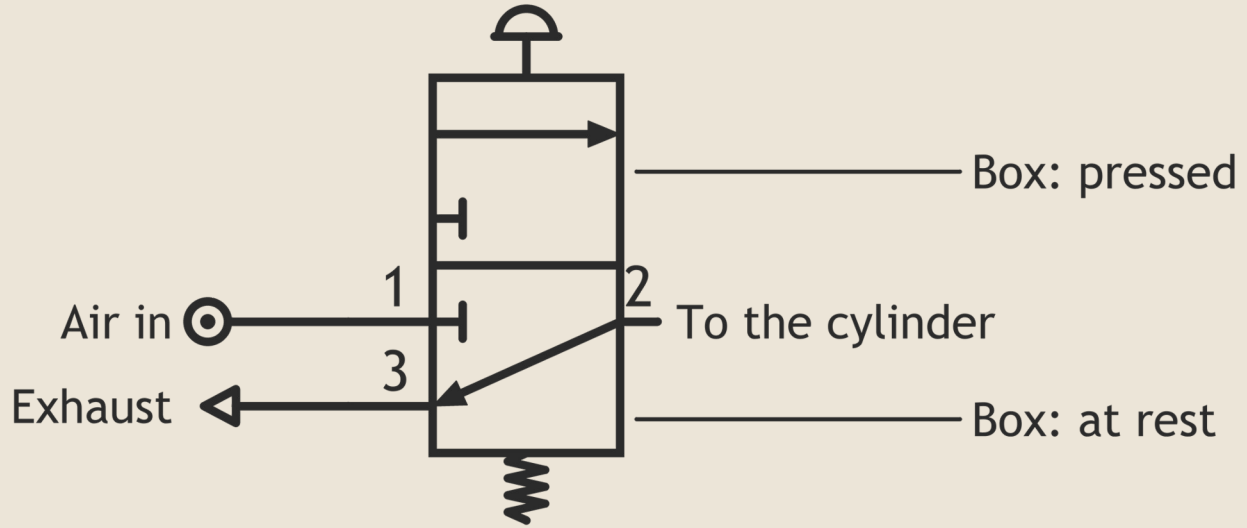
pressed and at rest

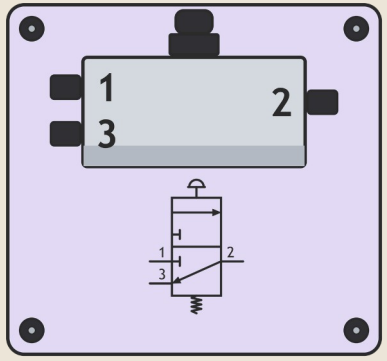
Ports

1 in, 2 out, 3 exhaust

3/2

3 ports and 2 states





The real valve. Its symbol is printed on the plate.

Parts of the Symbol (2 of 2): Inside the Boxes

Arrow

the path the air takes

T

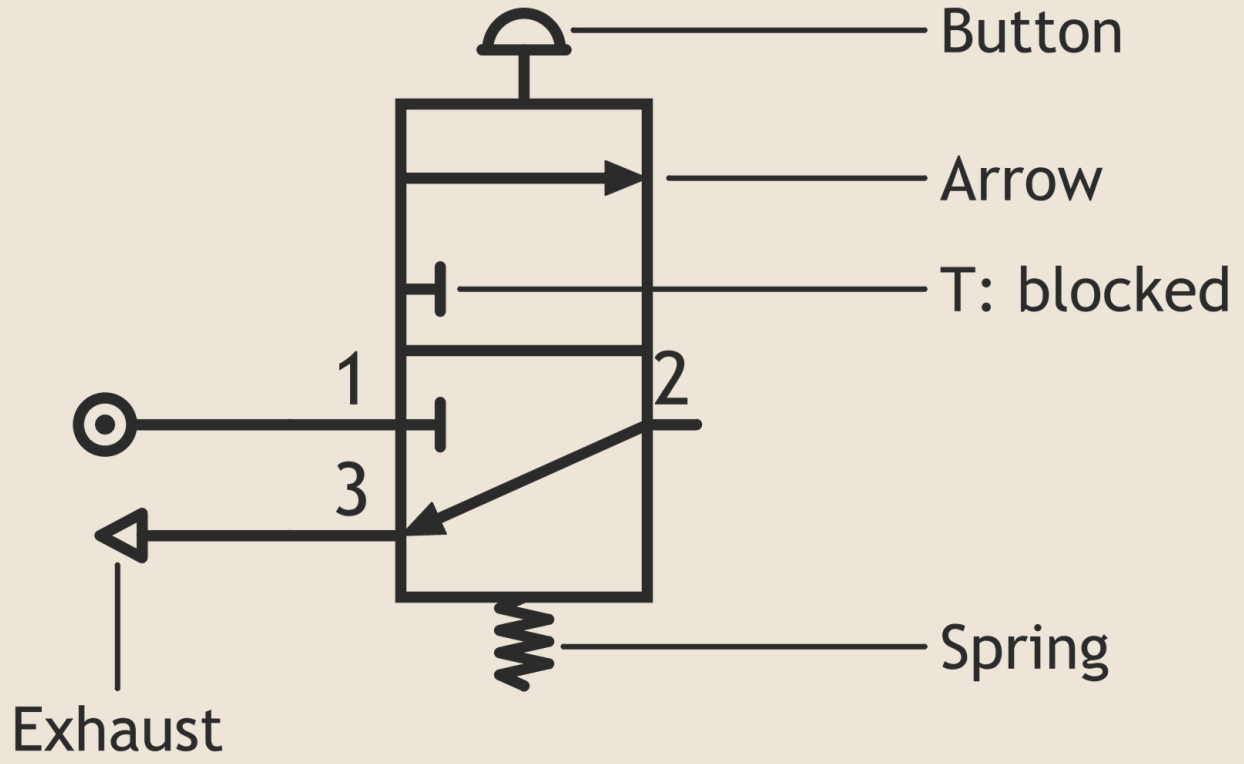
that port is blocked

Button

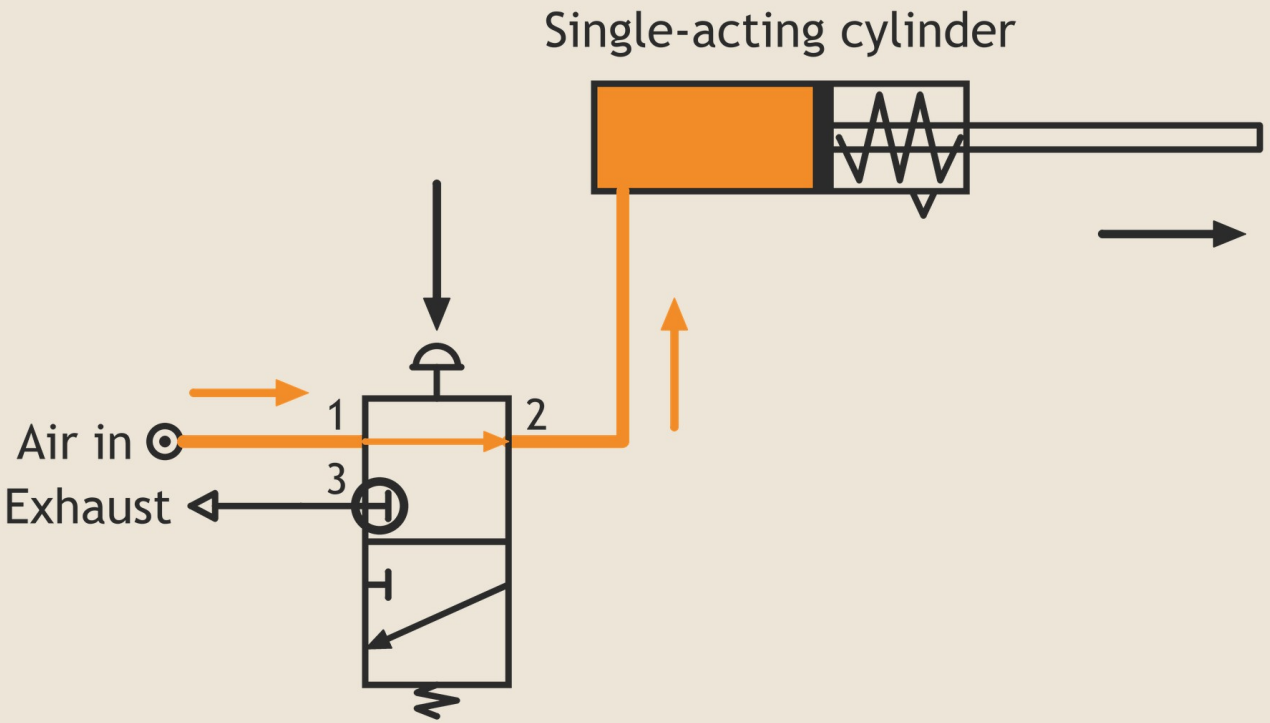
push it to change state

Spring

pushes the valve back



Worked Example: Press the Button



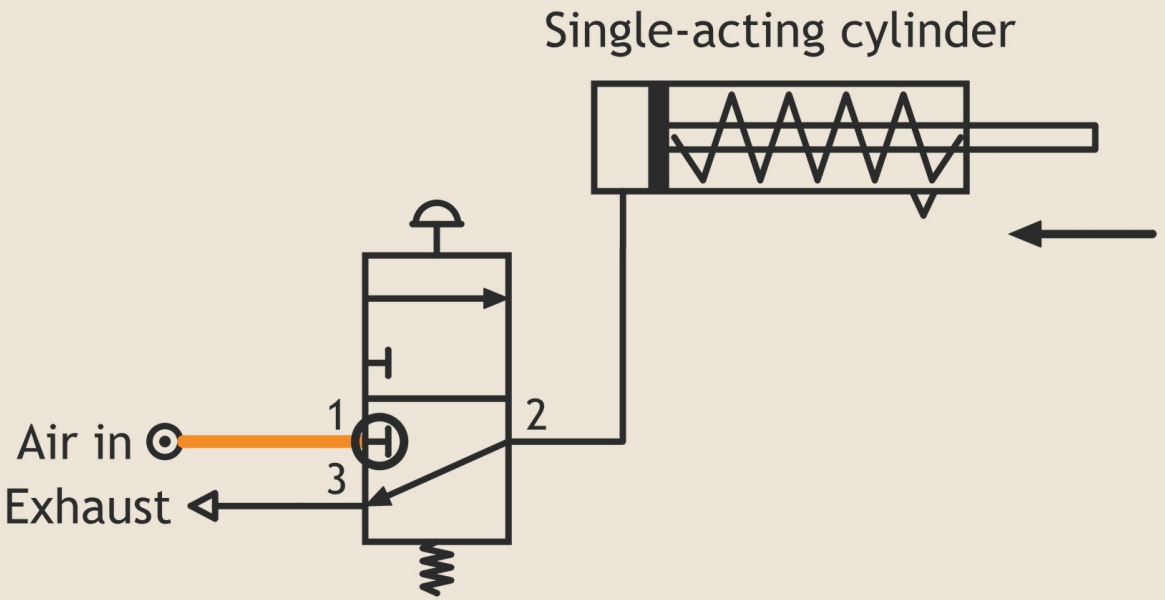
1 Push the button: both boxes slide down.

2 Port 3 is now blocked.

3 Air flows in port 1, out of port 2.

4 The rod moves out: the cylinder extends.

We Do: Let Go



1 Let go: the spring pushes the boxes up.

2 Port 1 is now blocked.

3 Air leaves: in port 2, out of port 3.

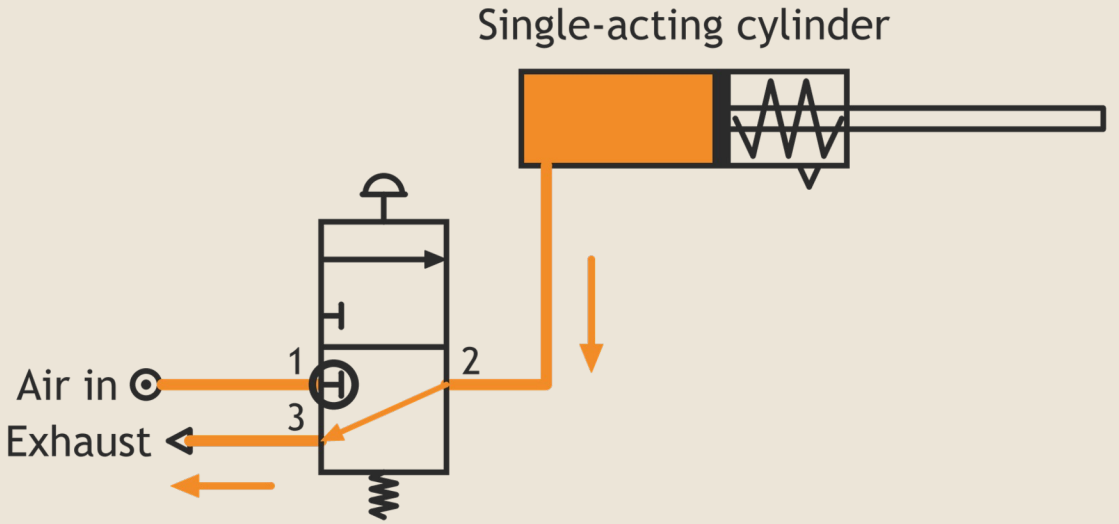
4 The rod moves in: the cylinder retracts.

Spot the Mistake

A pupil wrote: “When I let go, air goes into port 3 and pushes the rod back in.”

Find the mistake. **Correct** it.

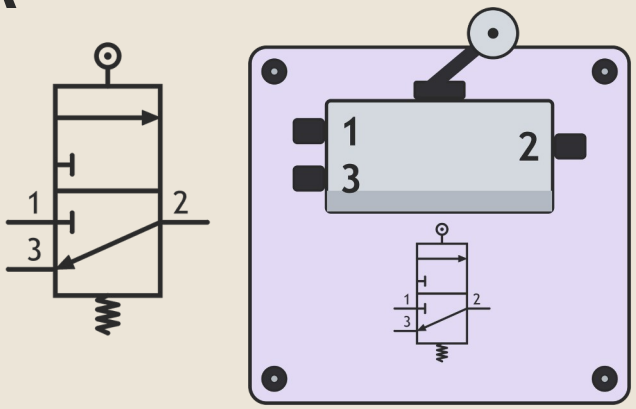
Air goes OUT of port 3. Port 1 is blocked. The spring pushes the rod in.



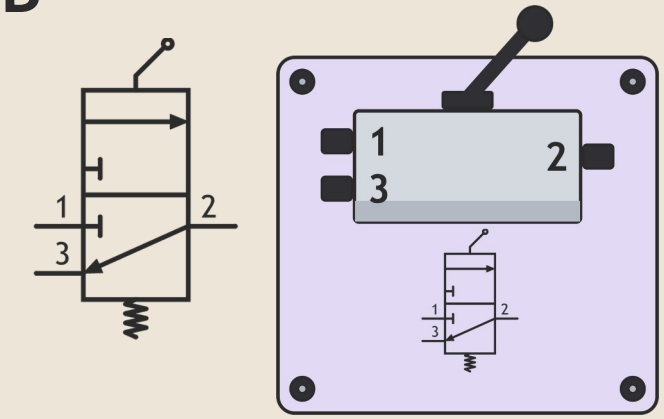
Show Me: Same Valve, Different Push

Name the **part on top** of valves A, B and C.

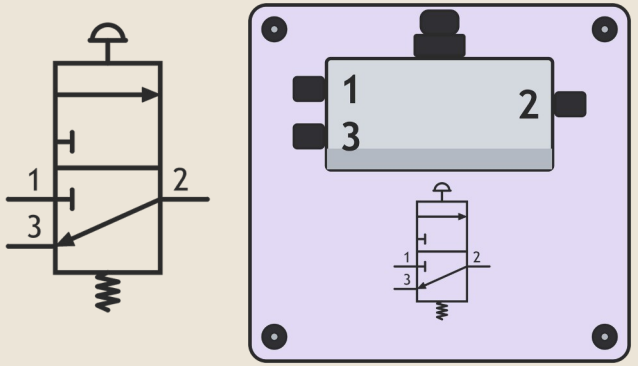
A



B



C



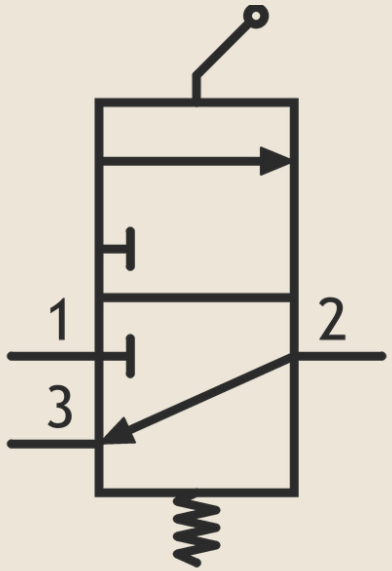
A Roller: a train presses it

B Lever: a hand moves it

C Button: a finger pushes it

Now You, Then Your Turn

Task 1 (B6A-L2-p5)



Answer Question 1 with a partner.
Answer Question 2 on your own.
Use the Let go steps to help.

5 min

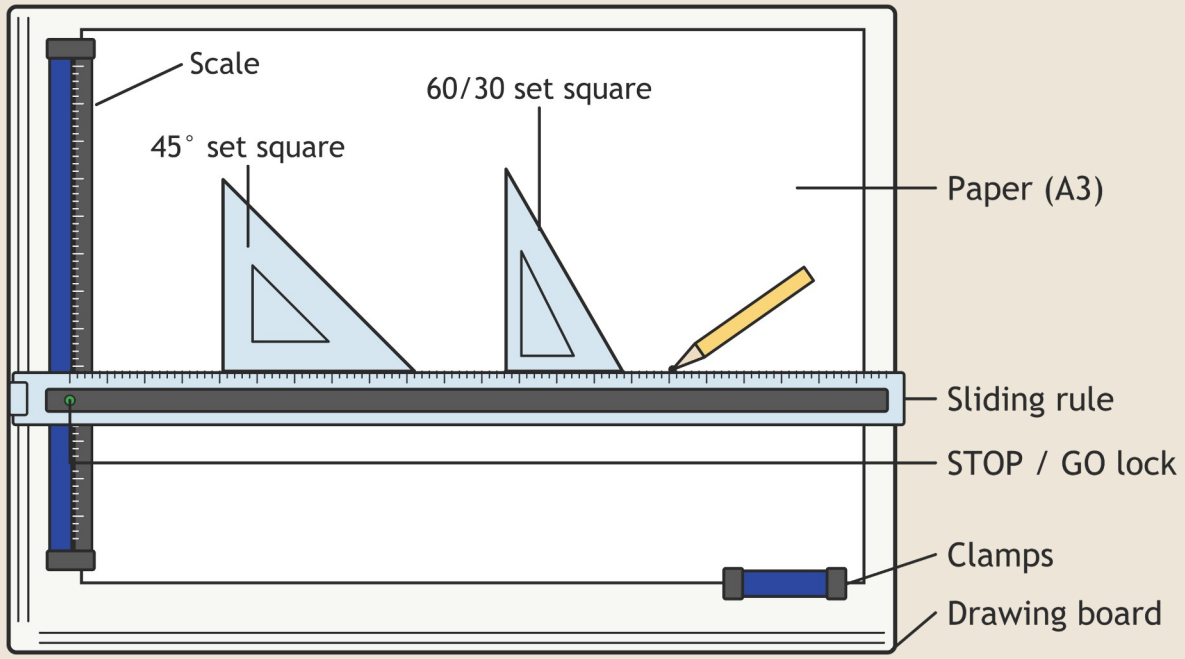
Answers: Task 1 (B6A-L2-p5)

Question	Answer
Complete line 2 (Q1)	Air goes in and pushes the piston.
Complete line 3 (Q1)	The rod moves out (it extends).
Describe: pressed (Q2)	The rod moves out.
Describe: train gone (Q2)	The spring pushes the rod back in.

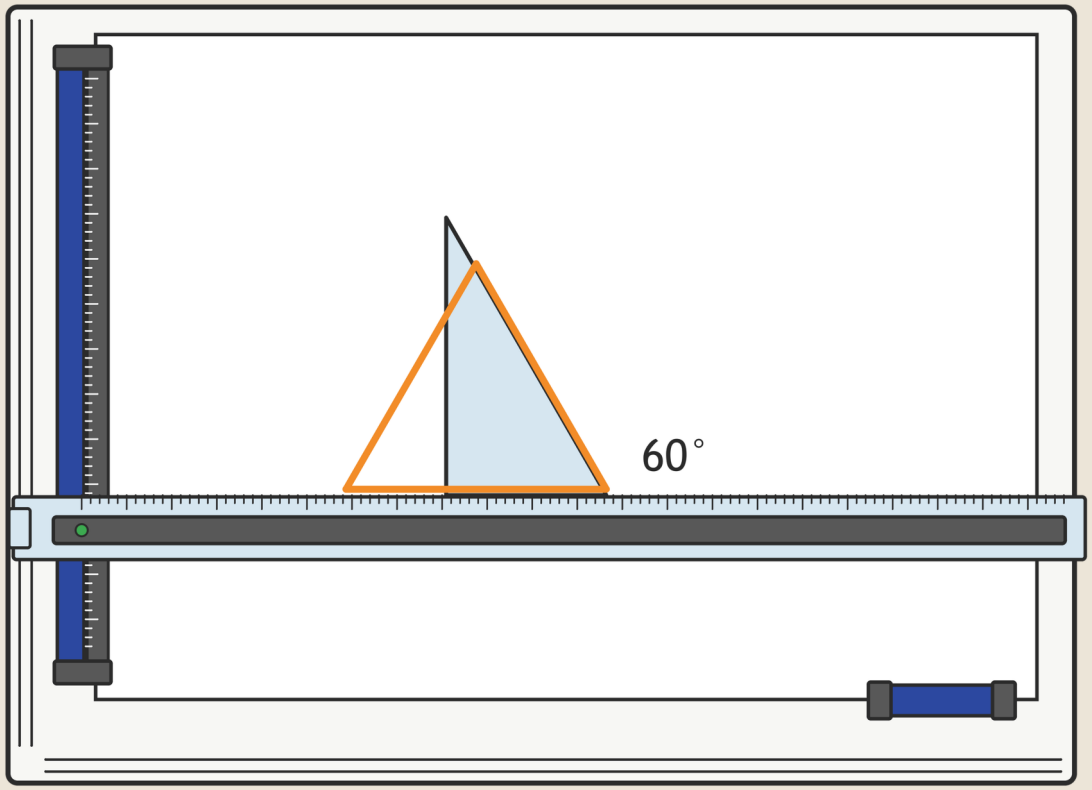
What to Collect

Collect one of each.

1 Drawing board
2 Plain A3 paper
3 45° set square
4 60/30 set square
5 Sharp pencil
6 Eraser



Using the Board



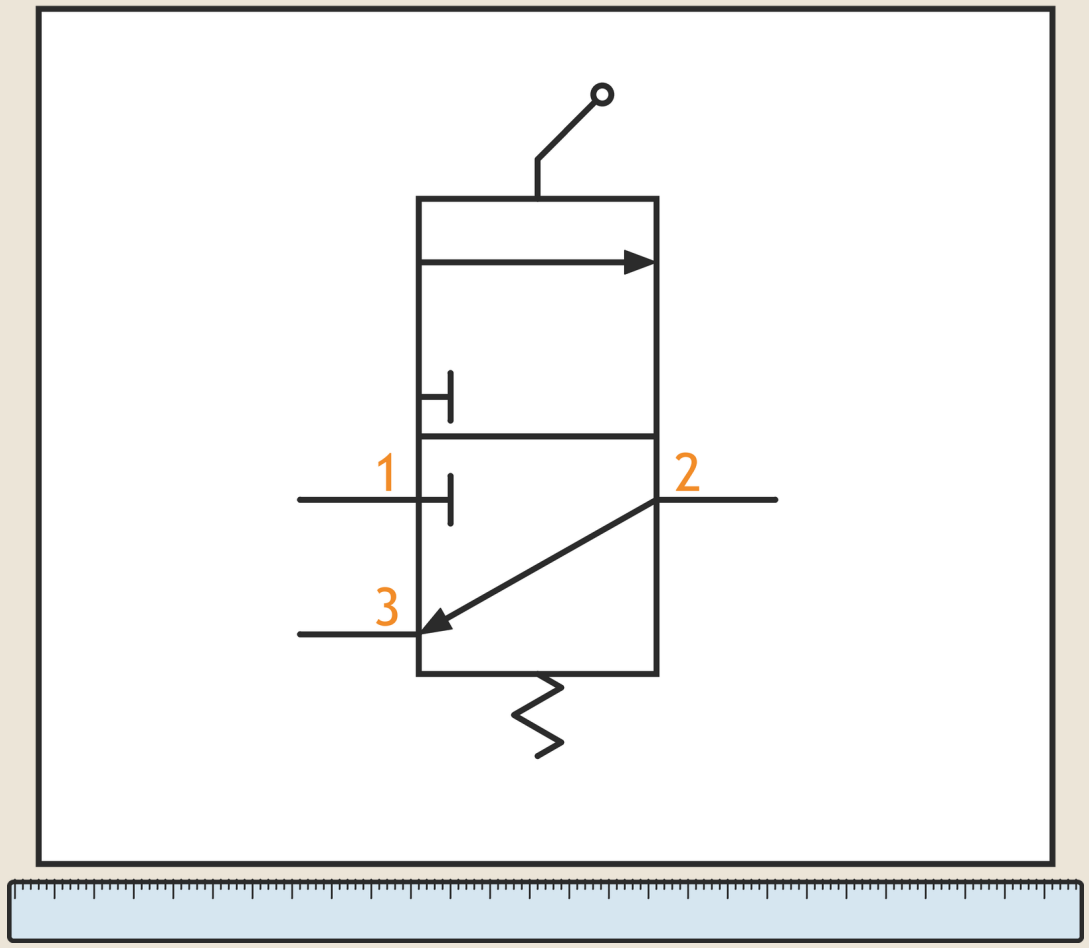
1 Clamp: paper in the bottom left corner.

2 Horizontal: pencil along the top of the rule.

3 Vertical: set square on the rule.

4 Angles: the 60° corner draws the exhaust.

Draw the Valve, Full Size



1 Two 30 mm squares, one on top.

2 Port lines, 15 mm long.

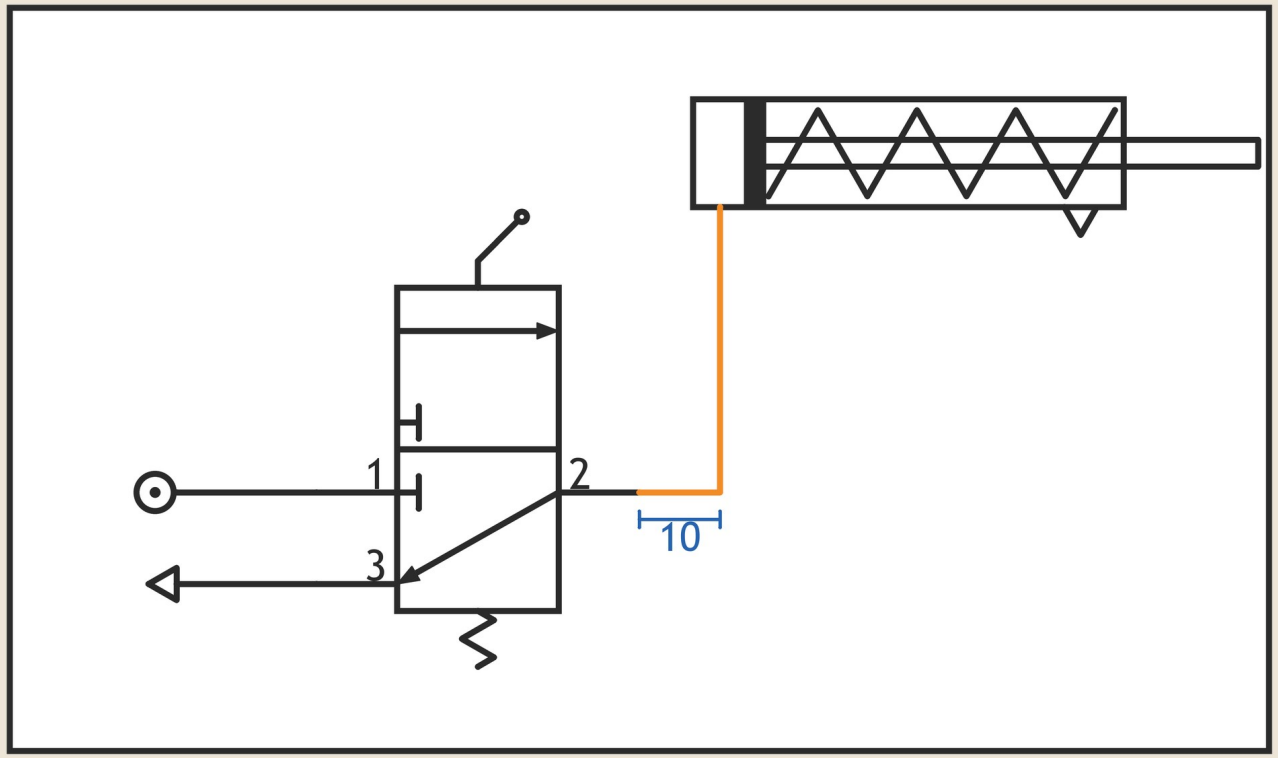
3 Straight arrow and two Ts.

4 Diagonal: 60/30 set square, 30° corner left.

5 Lever at 45°. Spring at 30°.

6 Number the ports 1, 2, 3.

Draw the Rest, Full Size



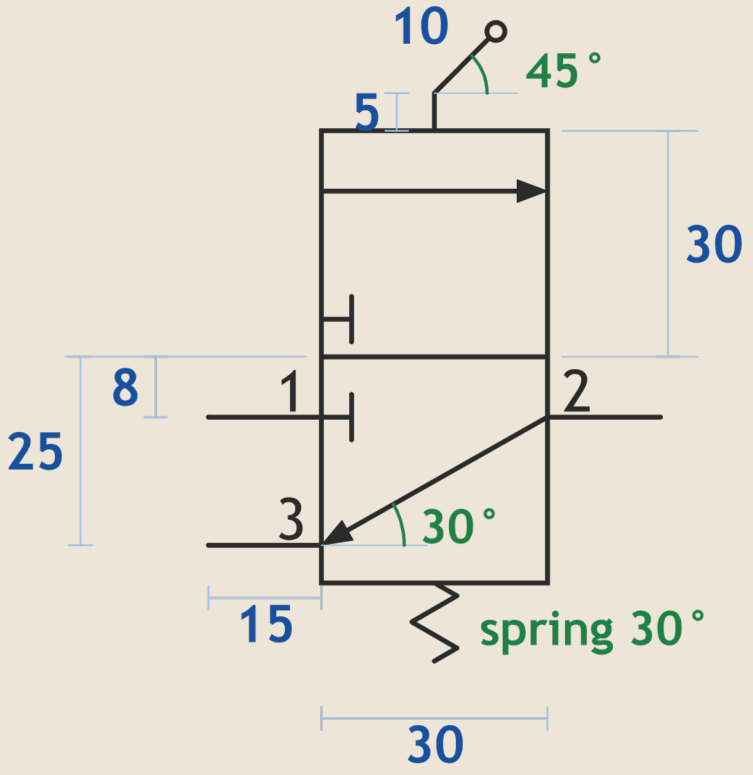
1 Cylinder 80 × 20 mm; rod 25 mm out.

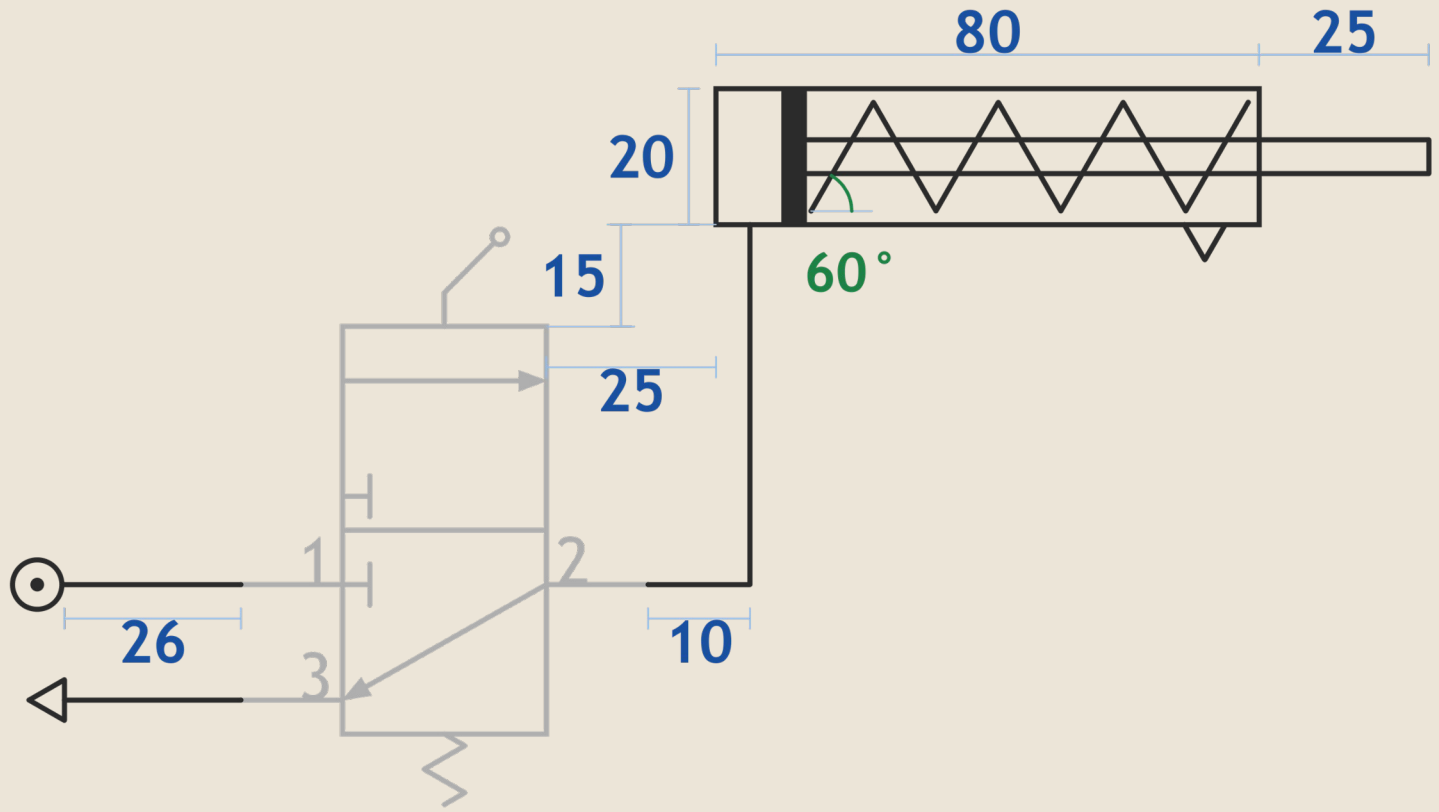
2 Spring at 60°; small triangle below.

3 Air in: 8 mm circle. Exhaust: 6 mm triangle.

4 Pipe: 10 mm across, then up.

Draw the Gate Circuit





Task 2 (B6A-L2-p5): **Draw** it full size on plain A3.
Small parts by eye. Pipes across or up only.

7 min

Check Your Drawing

Sizes

the same as the guide

Ports

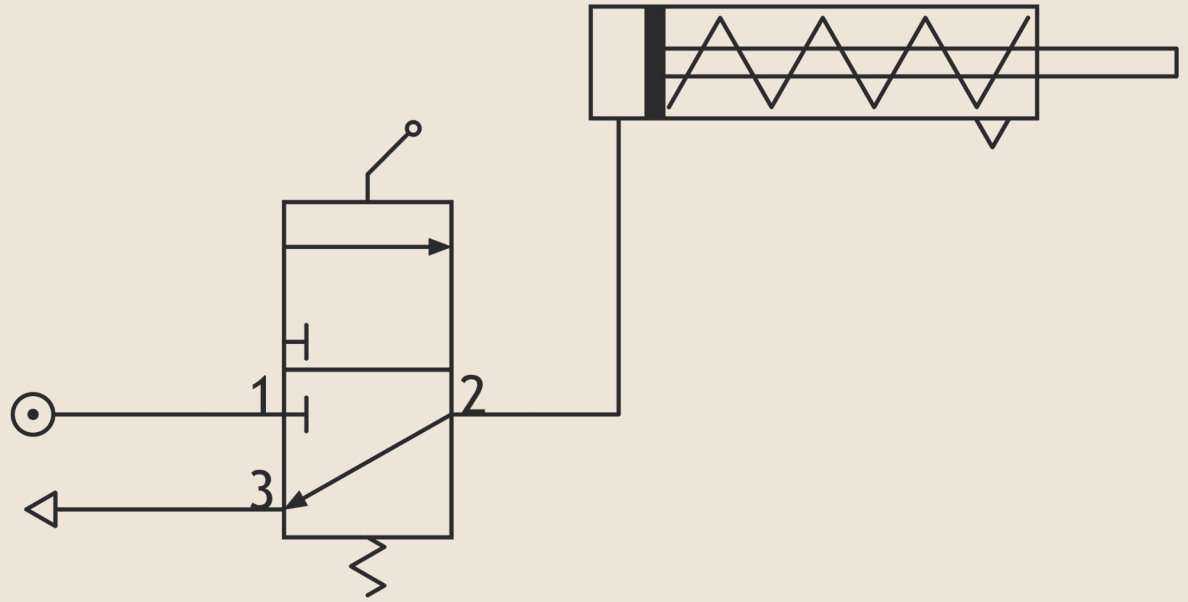
numbered 1, 2, 3

Exhaust

on port 3

Pipes

across or up only



Questions and Key Words

Tasks 5-6 (B6A-L2-p6)

Task 5: **Answer** Questions 3 and 4 on your own.
Task 6: **Complete** the key words.

4 min

Answers: Tasks 5-6 (B6A-L2-p6)

1	**Explain** why this valve is called a **3/2** valve. (Question 3)	It has 3 ports and 2 states
2	**State** what the **gauge** must read before you **change** a circuit. (Question 4)	0 bar
3	**Complete** key words line 1.	ports, states
4	**Complete** key words lines 2 and 3.	1, 2; exhaust

Plenary: Reading the Diagram

1. **State** how many **states** a **3/2 valve** has.

   Two states

2. **State** what a **T** inside a valve box **means**.

   That port is blocked

3. **State** which port is **blocked** when the **button is pressed**.

   Port 3

Plenary: Reading the Diagram

4 **State** what **pushes** the valve **back** when you **let go**.

The spring

5 **State** which **edge** of the rule you draw a **horizontal line** along.

The top edge of the rule

Look back — by the end of this lesson…



1. I can say what a **3/2 valve** does when it is **pressed** and **let go**
2. I can **draw** a circuit with **symbols**
3. I can **build** a circuit from a **bench card**

LOOKING AHEAD

Lesson 3: Control the Flow

Find out how to make the rod move **slowly**.

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Gate circuit: The gate circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 1 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Valve symbol and gate circuit pictures: The gate circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 1 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Real valve drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

What to Collect and Using the Board: Sources: drawing board pictures drawn by R Stewart from photos of the class boards.

R Stewart - October 2026